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Department of Mathematics
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Math 101 Final Exam Term 233

Original exam date: 15 August 2024. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The slope of the graph of the function $f(x) = \dfrac{3x+1}{(4x-3)^3}$ at the point $(1, 4)$ is

(a) 55

(b) 54

(c) -45

(d) 45

(e) -54

2. If $f(x) = \arctan(\sinh x)$, then $f'(\ln 2) =$

(a) $\dfrac{3}{4}$

(b) 2

(c) $\dfrac{4}{3}$

(d) $\dfrac{4}{5}$

(e) $\dfrac{5}{4}$

3. If $L(x)$ is the tangent line approximation to the graph of $f(x) = \log_2 x$ at the point $(2, 1)$, then $L(1) =$

(a) 0

(b) $1 + \dfrac{1}{\ln 2}$

(c) $1 - \dfrac{1}{2\ln 2}$

(d) $2 - \dfrac{1}{\ln 2}$

(e) $2 + \dfrac{1}{2\ln 2}$

4. $\lim_{x\to 0^+}(e^x + x)^{2/x} =$

(a) e^2

(b) e^4

(c) e^3

(d) e

(e) 1

5. If the function $f(x) = (x-1)^2(x+3)$ has a relative maximum at the point (a, b), then $a =$

(a) -1

(b) $\dfrac{3}{5}$

(c) $\dfrac{-5}{3}$

(d) $\dfrac{-3}{5}$

(e) $\dfrac{5}{3}$

6. If $f''(x) = \dfrac{2}{x^2}$, $f'(1) = 4$, and $f(1) = 3$, then $f(-1) =$

(a) -11

(b) -9

(c) 9

(d) -6

(e) 6

7. The equation of the normal line to the graph of the equation $\ln(x + y) = x$ at the point $(0, 1)$ is

(a) $y = 1$

(b) $y = 0$

(c) $y = x + 1$

(d) $x = 0$

(e) $x = 1$

8. The graph of the function $f(x) = \dfrac{2x^2}{3x^2+1}$ is concave upward on the interval

(a) $\left(-\dfrac{1}{3}, \dfrac{1}{3}\right)$

(b) $(-1, 1)$

(c) $\left(-\infty, -\dfrac{1}{3}\right)$

(d) $\left(-\dfrac{1}{3}, \infty\right)$

(e) $\left(\dfrac{1}{3}, \infty\right)$

9. Let x and y be two positive numbers such that their product is 147 and the sum of the first number plus three times the second number is a minimum. Then $x + y =$

(a) 30

(b) 26

(c) 34

(d) 28

(e) 32

10. The radius of a spherical ballon is measured as 8 inches, with a possible error of 0.02 inch. The percent error, approximately, in computing the volume of the sphere is, (Note, the volume of the sphere is $V = \dfrac{4}{3}\pi r^3$).

(a) $\dfrac{1}{4}\%$

(b) 2%

(c) 1%

(d) $\dfrac{1}{2}\%$

(e) $\dfrac{3}{4}\%$

11. If $y = \dfrac{(2x+1)^3(x^2-1)^2}{x+3}$, then $\dfrac{dy}{dx}$ at $x = 0$ equals

(a) $\dfrac{-17}{9}$

(b) 1

(c) $\dfrac{-9}{17}$

(d) $\dfrac{17}{9}$

(e) $\dfrac{9}{17}$

12. If $\lim_{x\to 0} \frac{a-\cos bx}{x^2} = 2$, where $b > 0$, then $a + b =$

(a) 4

(b) 2

(c) 0

(d) 1

(e) 3

13. If $x \sin y = y \cos x$ defines y as an implicit function of x, then $\dfrac{dy}{dx} =$

(a) $\dfrac{y \sin x + \sin y}{\cos x - x \cos y}$

(b) $\dfrac{y \cos x + \cos y}{\sin x - x \sin y}$

(c) $\dfrac{y \cos x - \cos y}{\sin x + x \sin y}$

(d) $\dfrac{y \sin x - \sin y}{\cos x + x \cos y}$

(e) $\dfrac{y \sin x}{x \cos y}$

14. The number of inflection points for the graph of the function $f(x) = x + 2\cos x$ over $[0, 2\pi]$ is

(a) 0

(b) 4

(c) 3

(d) 2

(e) 1

15. The function $f(x) = x + \dfrac{9}{x}$ is decreasing on the interval

(a) $(-9, 0)$

(b) $(-3, 0)$

(c) $(3, \infty)$

(d) $(-9, 9)$

(e) $(-\infty, -3)$

16. $\lim_{x\to\infty} \left(\frac{-4}{3+3e^{2x}} + \frac{6}{5+2e^{-4x}} \right) =$

(a) 0

(b) $\dfrac{2}{15}$

(c) $\dfrac{-4}{3}$

(d) $\dfrac{-12}{5}$

(e) $\dfrac{6}{5}$

17. If $f(x) = x \sin^2 x$, then $f''\left(\dfrac{\pi}{4}\right) =$

(a) $2 + \pi$

(b) π

(c) 2

(d) $2 - \pi$

(e) 2π

18. If $y = Ax + B$ is the slant asymptote for the graph of the function $f(x) = \dfrac{-x^2 - 4x - 7}{x + 3}$, then $A + B =$

(a) -1

(b) 2

(c) -2

(d) 0

(e) 1

19. Which one of the following functions is continuous on the entire real number line?

(a) $f(x) = \begin{cases} x^2 \sin \dfrac{1}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

(b) $f(x) = \tan x$

(c) $f(x) = \begin{cases} x \sin \dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

(d) $f(x) = \lfloor x \rfloor$

(e) $f(x) = \begin{cases} \sin \dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

20. $\lim_{x \to 0^+} \left(\ln(\sin x) - \frac{\cos^2 x}{x} \right) =$

(a) $-\infty$

(b) ∞

(c) 1

(d) 0

(e) -1

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	Quotient rule at $x=1$: $3-4\cdot 12=-45$.
2	**(d)**	$f'=\frac{1}{\cosh x}$, $\cosh(\ln 2)=\frac{5}{4}$, so $f'(\ln 2)=\frac{4}{5}$.
3	**(c)**	$L(x)=1+\frac{x-2}{2\ln 2}$, so $L(1)=1-\frac{1}{2\ln 2}$.
4	**(b)**	$\ln y=\frac{2\ln(e^x+x)}{x}\to 4$ by L'Hopital, so the limit is e^4.
5	**(c)**	$f'=(x-1)(3x+5)$; f' changes $+\to-$ at $x=-\frac{5}{3}$.
6	**(b)**	$f'=-\frac{2}{x}+6$, $f=-2\ln\lvert x\rvert+6x-3$, so $f(-1)=-9$.
7	**(d)**	$\frac{1+y'}{x+y}=1$ gives $y'(0,1)=0$; the normal is $x=0$.
8	**(a)**	$f''=\frac{4(1-9x^2)}{(3x^2+1)^3}>0$ iff $\lvert x\rvert<\frac{1}{3}$.
9	**(d)**	$S=x+\frac{441}{x}$, $x=21$, $y=7$, so $x+y=28$.
10	**(e)**	$\frac{dV}{V}=\frac{3\,dr}{r}=\frac{0.06}{8}=0.75\%=\frac{3}{4}\%$.
11	**(d)**	Log differentiation: $\frac{y'}{y}=6-\frac{1}{3}=\frac{17}{3}$, $y(0)=\frac{1}{3}$, so $y'(0)=\frac{17}{9}$.
12	**(e)**	$a=1$ forced; L'Hopital twice gives $\frac{b^2}{2}=2$, $b=2$; $a+b=3$.
13	**(a)**	$\sin y+xy'\cos y=y'\cos x-y\sin x$, so $y'=\frac{y\sin x+\sin y}{\cos x-x\cos y}$.
14	**(d)**	$f''=-2\cos x$ changes sign at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$: 2 points.
15	**(b)**	$f'=\frac{x^2-9}{x^2}<0$ for $-3<x<3$, $x\neq 0$; so $(-3,0)$.
16	**(e)**	First term $\to 0$, second $\to\frac{6}{5}$; limit $\frac{6}{5}$.
17	**(c)**	$f''=2\sin 2x+2x\cos 2x$, so $f''(\frac{\pi}{4})=2$.
18	**(c)**	Long division: $f=-x-1-\frac{4}{x+3}$, so $A=B=-1$, $A+B=-2$.
19	**(c)**	Squeeze: $x\sin\frac{1}{x}\to 0=f(0)$; the others have a break.
20	**(a)**	$\ln(\sin x)\to-\infty$ and $-\frac{\cos^2 x}{x}\to-\infty$, so the limit is $-\infty$.